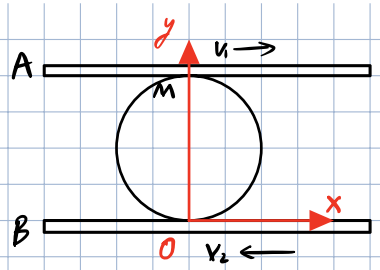
5.3




(i) $\vec{r}_S = \vec{r}_A + \frac{1}{\omega^2}\vec{\omega} \times \vec{v}_A$

$$\begin{cases} \vec{v}_M = \vec{v}_0 + \vec{\omega} \times \vec{r}_M = -v_2\vec{i} + \omega\vec{k} \times 2r\vec{j} = (-v_2 - 2\omega r)\vec{i} = v_1\vec{i} \\ \vec{v}_0 = \vec{v}_M + \vec{\omega} \times (-\vec{r}_M) = v_1\vec{i} + \omega\vec{k} \times (-2r\vec{j}) = (v_1 + 2\omega r)\vec{i} = -v_2\vec{i} \end{cases}$$

$$\Rightarrow \omega = -\frac{v_1+v_2}{2r}$$

$$\vec{r}_S = \vec{r}_M + \frac{1}{\omega^2}\vec{\omega} \times \vec{v}_A = 2r\vec{j} + \frac{1}{\omega^2}\omega\vec{k} \times v_1\vec{i} = 2r\left(1 - \frac{v_1}{v_1+v_2}\right)\vec{j}$$

$$= \frac{2rv_2}{v_1+v_2}\vec{j}$$

(ii) $v_M = \vec{v}_0 + \vec{\omega} \times \vec{r}_M$

$$\vec{a}_M = \vec{a}_0 + \dot{\vec{\omega}} \times \vec{r}_M + \vec{\omega} \times \dot{\vec{r}}_M = \vec{a}_0 + \vec{\omega} \times \dot{\vec{r}}_M$$

($\dot{\vec{\omega}} \times \vec{r}_M = 0$)

$$= \vec{\omega} \times (\vec{\omega} \times \vec{r}_M) = -\frac{(v_1+v_2)^2}{4r}\vec{j}$$

5.6.

$$J_{xx} = \int (y^2+z^2)\,dm$$

$$= \int \rho(y^2+z^2)\,dx\,dy\,dz$$

$$= \frac{m}{d^3}\int_{-\frac{d}{2}}^{\frac{d}{2}}dx\int_{-\frac{d}{2}}^{\frac{d}{2}}dy\int_{-\frac{d}{2}}^{\frac{d}{2}}(y^2+z^2)\,dz$$

$$= \frac{m}{d^3}\int_{-\frac{d}{2}}^{\frac{d}{2}}dx\int_{-\frac{d}{2}}^{\frac{d}{2}}\left(\frac{1}{3}z^3+zy^2\right)\Big|_{-\frac{d}{2}}^{\frac{d}{2}}dy = \frac{m}{d^3}\int_{-\frac{d}{2}}^{\frac{d}{2}}dx\int_{-\frac{d}{2}}^{\frac{d}{2}}\left(\frac{d^3}{12}+dy^2\right)dy$$

$$= \frac{1}{6}md^2$$

$$J_{yy} = J_{xx} = \frac{1}{6}md^2$$

$$J_{zz} = \int (x^2+y^2)\,dm = \frac{m}{d^3}\int (x^2+y^2)\,dx\,dy\,dz = \frac{m}{d^3}\left(\frac{1}{12}+\frac{1}{12}\right)d^5 = \frac{1}{6}md^2$$

$$J = \frac{1}{6}md^2\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$

惯量矩阵

平行轴定理.

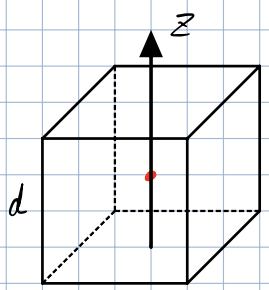
$$J_P = \frac{1}{6}md^2(1,0,0)\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1\\0\\0\end{pmatrix} + m\cdot\left(\frac{\sqrt{2}}{2}d\right)^2$$

转轴的方向余弦向量.

$$= \frac{2}{3}md^2$$

$$J_P = \left(0,\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2}\right)\frac{1}{6}md^2\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}0\\\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\end{pmatrix} + m\left(\frac{1}{2}d\right)^2$$

$$= \left(\frac{1}{6}+\frac{1}{4}\right)md^2 = \frac{5}{12}md^2$$



$$J_b = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right) \frac{1}{6} m d^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix} + 0.$$

$$= \frac{1}{3} \times 3 \times \frac{1}{6} m d^2 = \frac{1}{6} m d^2.$$

5.9

$$J_{xx} = J_{yy} = \int (x^2 + z^2)\, dm = \frac{m}{\pi r^2} \int (x^2 + z^2)\, dx\, dy\, dz$$

$$= \frac{m}{\pi r^2} \int x^2 dx dy = \frac{m}{\pi r^2} \int \rho^2 \cos^2\theta \rho d\rho d\theta = \frac{m}{\pi r^2} \int_0^{2\pi} d\theta \int_0^r \rho^3 \cos^2\theta\, d\rho$$

$$= \frac{m}{\pi r^2} \int_0^{2\pi} \frac{1}{4} r^4 \cos^2\theta\, d\theta = \frac{mr^2}{4\pi} \int_0^{2\pi} \frac{1-\cos 2\theta}{2} d\theta = \frac{mr^2}{4}$$

$$J_{zz} = \int (x^2 + y^2)\, dm = \int x^2 dm + \int y^2 dm = \frac{1}{2} m r^2$$

$$J = \frac{1}{4} m r^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$E = \frac{1}{2} \vec{\omega} \times \vec{J} \times \vec{\omega} + \frac{1}{2} m \vec{u}^2 \qquad \omega =$$

$$= \frac{mu^2}{32} (24 + \sin^2\alpha)$$

5.13

$$J = \int r^2 dm = 2\int_0^{\alpha} (2R\sin\alpha)^2 \sigma R d\alpha$$

$$= 4\sigma R^3 (\alpha_0 - \sin\alpha_0)$$

